

# Ryan Huang

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## TECHNICAL SKILLS

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**Programming Languages:** C/C++, Python, Bash, Java, MATLAB, R, SQL

**Firmware & Embedded Protocols:** FreeRTOS, Linux, JTAG, GDB, UART, I2C, SPI, TCP/IP, SNMP, ICMP

**Development Tools:** Git, Docker, Wireshark, Altium Designer, SolidWorks

## EDUCATION

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**The University of British Columbia**

Vancouver, BC

*Bachelor of Applied Science (BASc) in Biomedical Engineering*

*Expected Graduation: May 2028*

Relevant Courses: Control Systems, Circuits & EM, Biomedical Instrumentation, MEMS Devices, DSP & DIP

## WORK EXPERIENCE

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**Robert Bosch GmbH**

Sep. 2026 - Apr. 2027

*Hardware Engineering Intern*

*Reutlingen, Germany*

**Corinex**

Jan. 2026 - Aug. 2026

*Firmware Developer Co-op*

*Vancouver, BC*

- Investigated embedded firmware failures using JTAG/GDB and implemented 20+ bug fixes
- Programmed a flash utility tool using a JTAG peripheral with GDB scripts to write firmware images directly to SPI flash through memory-mapped registers
- Redesigned the shell logging system by implementing a lock-efficient static ring buffer, reducing mutex contention and improved concurrent logging performance by 90%
- Developed BusyBox shell scripts to monitor PHY speed, uptime, and LTE RSSI for embedded device validation
- Designed an optical port converter PCB in Altium Designer to interface embedded communication hardware
- Developed a Python automated OCPP validation framework that reduced manual testing time by 90%
- Supported deployment, validation, and compliance testing of broadband-over-power-line systems for smart grid and EV charging applications
- [More Information](#)

## PROJECT EXPERIENCE

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**Automated Finger Stretching Device for Quadriplegia**

Sep. 2025 - Sep. 2026

*Electrical & Embedded Team Lead on UBC BEST - ENABLE*

*Vancouver, BC*

- Led the electrical and embedded systems development of a multi-motor finger stretching device for a quadriplegic client, translating design specifications into control system design
- Designed the embedded architecture by programming an Arduino multi-motor control framework, assembling the electrical system, and reducing wiring complexity for scalable actuator control
- Co-authored a peer-reviewed research abstract published in the *Gerontechnology* Journal and presented the project at the International Society for Gerontechnology Conference
- [Publication Link](#), [Project Details](#)

**Embedded Music Controller**

- Developed an Arduino music controller that directly accesses Spotify to display current song and artist along with volume and playback control
- Implemented bidirectional serial communication between Arduino firmware and a Python script with robust encoding and decoding, achieving perfect transmission reliability over 200+ test cycles
- Integrated the Spotify Web API using OAuth authentication to securely synchronize playback state between the controller and Spotify in real time
- [More Projects & Highlights](#)

## AWARDS

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**UBC Elizabeth and Leslie Gould Entrance Scholarship for Engineering**

Merit-based scholarship renewed annually for academic excellence, leadership, and co-curricular achievement in the BASc in Engineering program.

**Hobbies:** piano, music curation, basketball, ultimate frisbee, prototyping